RADIO GALAXY

M87

VIRGO

CATALOG NUMBERS M87, NGC 4486

SHAPE E1 giant elliptical

DISTANCE 60 million light-years

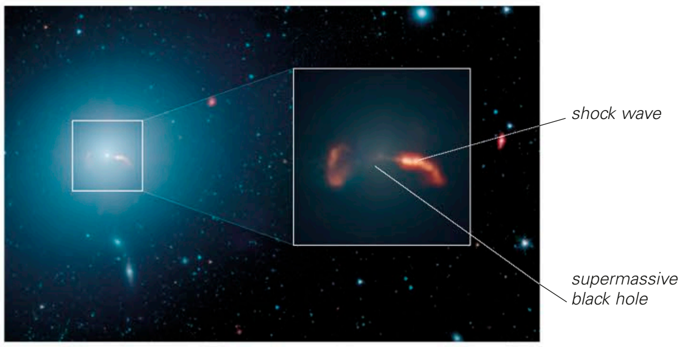
DIAMETER 120,000 light-years

MAGNITUDE 8.6

Lying at the heart of the Virgo Galaxy Cluster (see p.329), M87 is the closest example of a giant elliptical galaxy—a class of galaxy often found at the cores of old galaxy clusters. This huge ball of stars seems to have roughly the same diameter as the Milky Way but contains many more stars distributed across its spherical structure—probably several trillion. Long-exposure photographs reveal an extensive halo of more loosely scattered stars in a more elongated shape. The galaxy also has an unrivaled collection of globular star clusters in orbit—some astronomers estimate as many as 15,000 such groups.

What is more, M87 is an active galaxy—its location coincides with the Virgo A radio source and with a strong source of X-rays. There is even a sign of this activity that is visible at optical wavelengths in the form of a long, narrow jet of material being blasted from its interior. In 2019, the Event Horizon Telescope successfully imaged M87's enormous central black hole (see p.26).

GALACTIC ERUPTION
Infrared images from the Spitzer Space Telescope show shock waves around Messier 87's high-energy jets.



TYPE-II SEYFERT GALAXY

Fried Egg Galaxy

PEGASUS

CATALOG NUMBER NGC 7742

SHAPE Sb spiral

DISTANCE 72 million light-years

DIAMETER 36,000 light-years

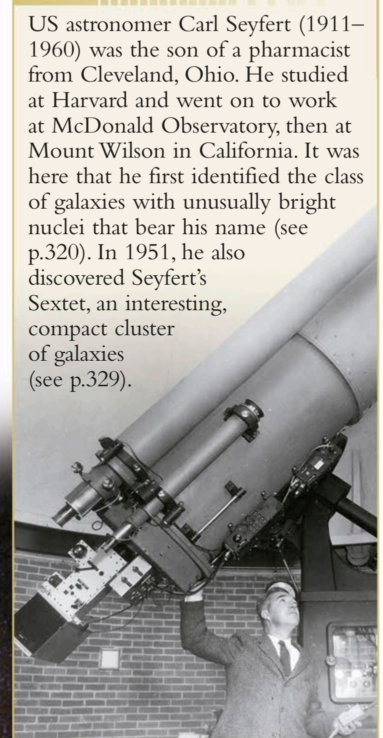
MAGNITUDE 11.6

The small spiral galaxy NGC 7742 resembles a fried egg because of the intense yellow glow from its core. The core is much brighter than is usual for a galaxy of this size, because this is a Seyfert galaxy with a moderately active core. Seyferts emit radiation across a broad band of wavelengths—NGC 7742 is a type-II—a galaxy that is brightest in infrared light.

CELESTIAL EGG

CARL SEYFERT

US astronomer Carl Seyfert (1911–1960) was the son of a pharmacist from Cleveland, Ohio. He studied at Harvard and went on to work at McDonald Observatory, then at Mount Wilson in California. It was here that he first identified the class of galaxies with unusually bright nuclei that bear his name (see p.320). In 1951, he also discovered Seyfert's Sextet, an interesting, compact cluster of galaxies (see p.329).



SEYFERT'S OBSERVATORY
At Nashville, Seyfert found time to give public lectures, as well as raising support and supervising the construction of the Arthur J. Dyer Observatory (above).

RADIO GALAXY

NGC 4261

VIRGO

CATALOG NUMBER NGC 4261

SHAPE E1 elliptical

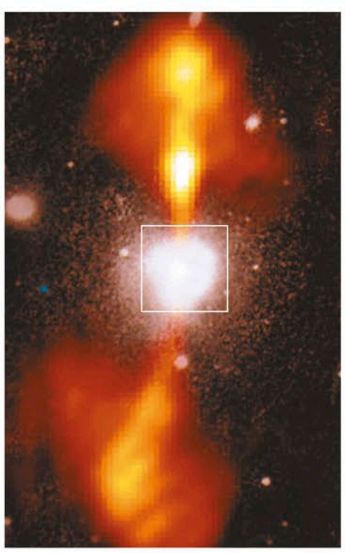
DISTANCE 100 million light-years

DIAMETER 60,000 light-years

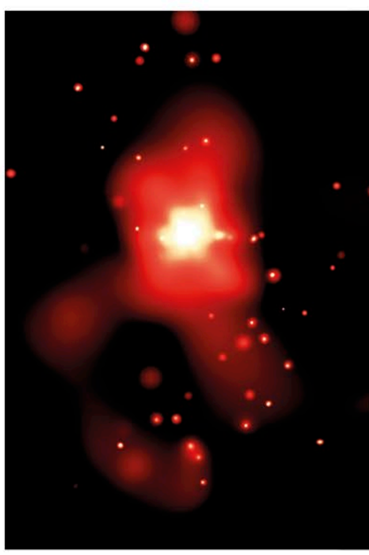
MAGNITUDE 10.3

The elliptical galaxy NGC 4261 lies at the center of two great lobes of radio emission measuring 150,000 light-years from tip to tip. In many ways a typical radio galaxy, it is also one of the few active elliptical galaxies to have revealed its internal structure to astronomers. Infrared images from the Hubble Space Telescope pierced the obscuring clouds of stars to reveal an unexpectedly dense disk of dusty material, apparently spiraling onto the galaxy's central black hole.

Most elliptical galaxies are thought to be relatively dust-free, so where did the material in NGC 4261 come from? The most likely answer is that the elliptical galaxy has merged with a spiral in its relatively recent history. The spiral's individual stars have now become indistinguishable from the stars that were originally part of the elliptical galaxy, but the ghostly outline of the galaxy's gas and dust remains.



RADIATING PLUMES
Combining optical and radio images of NGC 4261 reveals its full extent. The visible part of the galaxy is the white blob in the center, while the orange plumes mark the radio-emitting regions.

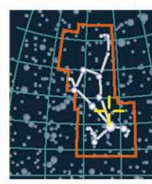


ANCIENT REMAINS
A Chandra X-ray image of the galaxy's core reveals dozens of black holes and neutron stars around the supermassive central black hole, perhaps formed during its recent collision.

DUST WHIRLPOOL
The Hubble Space Telescope's close-up image of the core reveals a dusty spiral of matter within a ring of glowing outer clouds. A distinct cone shows where matter is being flung off from the active galactic nucleus into the radio lobes.

TYPE-I SEYFERT GALAXY

NGC 5548



BOÖTES

CATALOG NUMBER NGC 5548

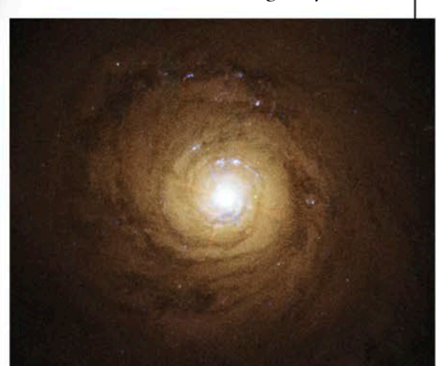
SHAPE Sb spiral

DISTANCE 220 million light-years

DIAMETER 100,000 light-years

MAGNITUDE 10.5

NGC 5548 is a type-I Seyfert galaxy—that is, a Seyfert that emits more ultraviolet and X-ray radiation than visible light. Like all Seyferts, it has a bright, compact core but, unlike the Fried Egg Galaxy (see above), its core is an intense blue-white. Using the Chandra X-ray Telescope, astronomers have detected an envelope of warm gas expanding around the core. The gas eventually forms two lobes of weak radio emission around the galaxy.



HUBBLE IMAGE OF NGC 5548